

World-Model Collapse as a Phase Transition

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Abstract

Water looks unchanged as it warms, then at a critical point it boils. We ask whether long-horizon language agents show an analogous transition in their implicit world models. In some parameter settings, changing state load by a small amount, or adding a single step of horizon, leaves behavior nearly unchanged; near a critical boundary, the same small change causes a sudden world collapse. We study this effect in a deterministic task family with exact per-step gold state. A large grid search over state cardinality, dependency density, horizon, branching, observation mode, and mutation rate reveals a phase diagram: a solved plateau, a narrow transition band, and a collapse floor. Per-step traces show the mechanism: world-state fidelity fails before action validity, so the agent is not merely choosing a bad action; it is acting from a corrupted world. Stronger models translate the critical boundary but do not remove the qualitative transition. These results make world-model collapse a measurable bottleneck for long-horizon agents. Code is available at <https://github.com/Hik289/world-model-collapse.git>.

1 Introduction

Many systems look stable until a control parameter crosses a critical value. Water can be heated from 90°C to 99°C without changing phase, but near 100°C a small increase produces boiling. We argue that long-horizon language agents can fail in the same qualitative way. An agent may track a task, update memory, and choose plausible actions for many steps; then a small increase in state load, dependency density, or horizon pushes the implicit world model past a critical point. The result is not gradual degradation, but world collapse: the agent continues to reason fluently while the represented world it reasons over has become wrong.

This analogy is useful because phase transitions are not defined only by dramatic outcomes; they are defined by the geometry of a response surface.

Statistical physics distinguishes control parameters, order parameters, critical regions, finite-size crossovers, and precursors to collapse (Stanley and Ahlers, 1973; Goldenfeld, 2018; Sethna, 2021). We use the same operational language without claiming a thermodynamic limit. In our setting, state load and dependency density are control parameters, task success is the order parameter, and world-state fidelity is the precursor. The question is whether the observed surface looks like smooth drift or like a finite-grid phase transition.

This view differs from the usual drift story. In ReAct-style and Reflexion-style agents (Yao et al., 2023b; Shinn et al., 2023), search scaffolds such as Tree-of-Thoughts and Graph-of-Thoughts (Yao et al., 2023a; Besta et al., 2024), and large agent benchmarks (Shridhar et al., 2021; Wang et al., 2022; Yao et al., 2022; Zhou et al., 2024; Liu et al., 2023b; Mialon et al., 2023; Jimenez et al., 2023; Xie et al., 2024a; Yao et al., 2024; Mazaheri and Mazaheri, 2026), failure is often interpreted as smooth accumulation of local errors. If that is the whole story, more search, more self-checking, or a longer horizon should often help. If the agent crosses a representational phase boundary, those tools arrive too late unless they preserve the world state itself. A final success score cannot distinguish these cases; the agent may make its first invalid move only after its internal world has already collapsed.

We therefore stress the world model directly. We separate state cardinality, SC, the number of entities that must remain jointly addressable, from dependency density, DD, the number of preconditions that bind each subgoal to the current state. We then run dense grid searches over these axes and over secondary controls such as horizon, branching, observation mode, and mutation rate. Smooth drift predicts gradual surfaces. A phase transition predicts a different geometry: a solved plateau, a narrow critical band, and a collapse floor. It also

predicts a temporal precursor: world-state fidelity should fail before action validity. Figure 1 summarizes the phase-diagram view, and the instrumented agent loop in Figure 2 measures the ordering.

We test the story in StatefulPuzzle, a deterministic environment with exact per-step gold state. This control lets us distinguish an agent that chooses a bad action from an agent that no longer represents the world correctly. The confirmatory grid in Figure 3 exposes a sharp boundary in the (SC, DD) plane, with the cell values reported in Table 2 and the one-dimensional cross-sections in Figure 4. The fine scan in Figure 7 localizes the critical point near $SC^* \approx 13.5$ for the main setting. The cross-model comparison in Figure 5 and Table 8 shows that stronger models translate the boundary rather than erasing it. Figure 6 and Table 4 check that the collapse is not a disguised effect of horizon, branching, observation noise, or mutation rate alone.

The paper follows that chain of evidence. We first build a controlled phase diagram, then use per-step traces to identify world-state failure as the precursor, then perturb alternative axes to rule out simpler explanations. The result is not an agent leaderboard. It is a measurement of where a model-agent pair loses the world it is trying to reason over.

Contributions. First, we formulate long-horizon agent collapse as a two-dimensional phase-transition problem in world-model capacity. Second, we introduce a controlled evaluation that separates state load from dependency load while logging world-state accuracy and action validity at every step. Third, we show that world-state collapse is the temporal precursor to plan collapse, turning an apparent action-selection error into a representational failure. Fourth, we demonstrate that model capability shifts the boundary and that secondary stress axes modulate, but do not replace, the governing (SC, DD) structure.

2 Related Work

Agent benchmarks and failure analyses. Interactive benchmarks have made language-agent failure visible across text worlds, web tasks, software engineering, travel planning, tool use, and general-assistant settings (Shridhar et al., 2021; Wang et al., 2022; Yao et al., 2022; Zhou et al., 2024; Liu et al., 2023b; Mialon et al., 2023; Jimenez et al., 2023; Yang et al., 2024; Xie et al., 2024a; Yao et al., 2024; Mazaheri and Mazaheri, 2026; Ding

et al., 2026). Recent evaluation critiques similarly warn that outcome-only benchmarks can conflate accuracy, cost, harness effects, and reproducibility (Kapoor et al., 2025). Their strength is ecological breadth; their weakness for our question is that final success typically confounds state size, dependency complexity, observation noise, horizon, and interface effects. Tool-use benchmarks and training sets add realistic API structure (Schick et al., 2023; Qin et al., 2023; Guo et al., 2024), while recent controllable planning data targets verifiability (Zhao et al., 2026). We take the complementary route: sacrifice realism to obtain exact per-step state and orthogonal stress axes, which are necessary to test the *geometry* of failure.

World models in LLM agents. The world-model framing has a long lineage (Ha and Schmidhuber, 2018; LeCun, 2022; Hafner et al., 2021, 2023; Bruce et al., 2024). LLM agents hold a world model only *implicitly*, inside the in-context state representation maintained by the prompt and memory. Recent probes of this implicit state (Chen et al., 2023; Hou et al., 2026; Zhu et al., 2026; Samiei et al., 2025; Chao et al., 2026) and world-model planning benchmarks (Chen et al., 2025) study whether models can maintain, revise, or use state. Cognitive-science critiques (Mahowald et al., 2024) ask what is missing architecturally; we ask *when it breaks*.

Planning failures in LLMs. Kambhampati (2024); Kambhampati et al. (2024) argue LLMs are approximate retrievers, not planners; the Plan-Bench line (Valmeekam et al., 2022, 2023, 2024; Stechly et al., 2024) documents sharp drops with problem complexity; hybrids externalise the planner (Liu et al., 2023a; Silver et al., 2024; Xie et al., 2024b). Search-based prompting improves some deliberative tasks by expanding the inference tree (Yao et al., 2023a; Besta et al., 2024), but it does not by itself explain when the state representation supporting the search collapses. We localise the boundary on two specific axes and identify its temporal order: world state first, plan validity second.

Memory and context budget. Memory taxonomies (Wang et al., 2024; Park et al., 2023; Sumers et al., 2024; Packer et al., 2024) and context-budget accounts of failure (Liu et al., 2024; Chung et al., 2025; Zhou et al., 2025; Luo et al., 2025; Ponnusamy et al., 2025; Fang et al., 2026)

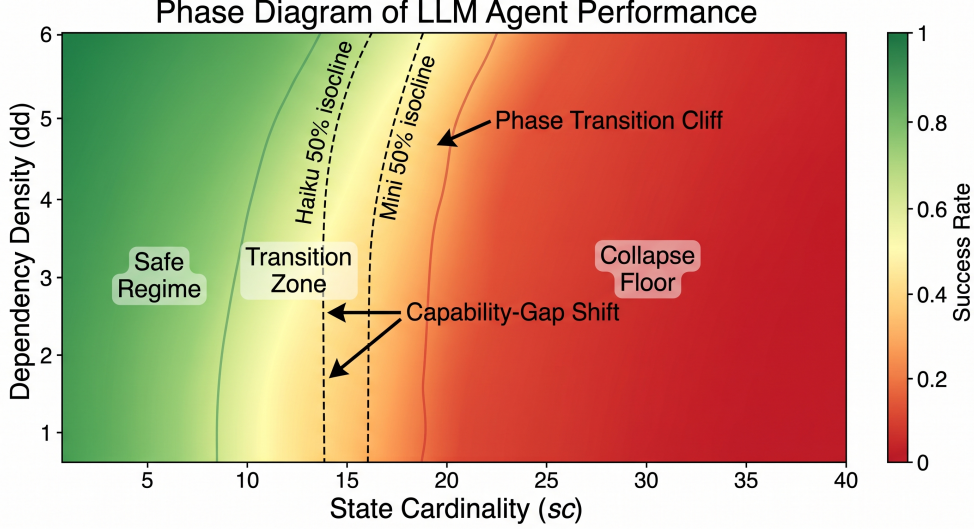


Figure 1: Conceptual phase diagram for world-model collapse. The (sc, dd) plane partitions into a solved regime, a narrow transition zone, and a collapse floor. A stronger or better-supported model shifts the boundary rightward, but the qualitative shape remains. The figure is a mechanistic summary, not an additional data plot.

attribute collapse to budget exhaustion, evidence pruning, or dynamic context synthesis. Revisitable and structured memory systems try to counter that loss (Shi et al., 2025; Arslan, 2026). Our T -axis ablation discriminates these mechanisms from a pure horizon account: the cliff appears while episodes remain well below the available context budget.

Phase transitions in neural networks. Classical statistical physics studies how macroscopic behavior changes abruptly as a control parameter crosses a critical point, with the transition described through order parameters, critical regions, finite-size effects, and scaling laws (Stanley and Ahlers, 1973; Goldenfeld, 2018; Sethna, 2021). This vocabulary has also shaped the analysis of learning systems, where smooth scaling laws coexist with sharp qualitative changes. Smooth scaling laws (Kaplan et al., 2020) coexist with sharp phenomena: emergent abilities (Wei et al., 2022) with the metric-artifact caveat (Schaeffer et al., 2023), grokking (Power et al., 2022; Nanda et al., 2023), induction-head formation (Olsson et al., 2022), double descent (Belkin et al., 2019; Nakkiran et al., 2020), and statistical-mechanics treatments (Mei et al., 2018; Bahri et al., 2020; Saxe et al., 2019; Roberts et al., 2022). Nakaishi et al. (2024) extend this lineage to a critical phase transition in static LLM output quality vs. decoding temperature. We extend it further to *sequential agent planning dynamics* with task success as the order parameter and a two-dimensional task-side control.

Gradual drift as the canonical baseline. The most direct contrast is the canonical-path-deviation framework of Lee (2026), which predicts a smooth sigmoid in any stress axis. We adopt that smoothness prediction as the canonical drift baseline. Drift exists, but it cannot explain the plateau-boundary-floor geometry that appears under two-dimensional structural stress.

3 Formal Framework

Let $\mathcal{D}_{s,d,z}$ denote a distribution over finite episodes. The structural coordinates are state cardinality $s \in \mathbb{N}$ and dependency density $d \in \mathbb{N}$. The nuisance vector z fixes horizon, branching, observation mode, mutation rate, and all other factors not under study. Each episode has gold world states $W_{0:T}^*$ generated by a deterministic simulator,

$$W_{t+1}^* = F(W_t^*, a_t; x), \quad x \sim \mathcal{D}_{s,d,z}.$$

The agent maintains an explicit working state $\widehat{W}_t$ and chooses $a_t = \pi_\theta(H_t, \widehat{W}_t)$ from the interaction history H_t . Final success is the Bernoulli variable $Y = \mathbf{1}\{G(W_T^*) = 1\}$, where G is the gold goal predicate. The order parameter is therefore

$$p_\theta(s, d; z) = \mathbb{E}_{x \sim \mathcal{D}_{s,d,z}, \pi_\theta}[Y].$$

Definition 1 (Finite-grid abrupt transition). *Fix z , finite grids $\mathcal{G}_s, \mathcal{G}_d$, a cliff margin $\delta > 0$, and plateau/floor thresholds $0 < \alpha < \beta < 1$. The observed surface has an abrupt transition if there*

exist adjacent $s_i < s_{i+1}$ and some $d \in \mathcal{G}_d$ such that

$$\hat{p}_\theta(s_i, d; z) - \hat{p}_\theta(s_{i+1}, d; z) \geq \delta,$$

and the same grid contains nonempty regimes with $\hat{p}_\theta \geq \beta$ and $\hat{p}_\theta \leq \alpha$. For fixed d , the operational crossing is

$$s_\theta^*(d; z) = \inf\{s : p_\theta(s, d; z) \leq 1/2\}.$$

The mechanism is defined at the trajectory level. Let $A_t = \mathbf{1}\{a_t \text{ is valid in } W_t^*\}$ and let $R_t = \rho(\widehat{W}_t, W_t^*)$ be a gold-scored world-state fidelity metric. For a window $h = 5$,

$$\tau_W = \inf\left\{t : h^{-1} \sum_{j=0}^{h-1} R_{t-j} < 1/2\right\},$$

$$\tau_A = \inf\{t \geq 1 : A_t = 0\}.$$

A representational-collapse account predicts $\tau_W < \tau_A$ on failed episodes: the world representation crosses its failure threshold before the first invalid action. The appendix gives the elementary grid-bracketing proof under monotonicity. The claim in the main text is finite-grid and operational, not a thermodynamic-limit statement.

4 Method

To isolate structural drivers of collapse from confounders such as horizon length, interface noise, or model size, we use controlled rule environments with exact per-step gold state. The design goal is not to approximate a deployed web task; it is to make the failure surface observable. This follows the controllable-benchmark tradition in planning and agent evaluation (Valmeekam et al., 2022; Zhao et al., 2026; Fang et al., 2026) while adopting the reproducibility concerns raised by recent agent-evaluation critiques (Kapoor et al., 2025).

Environments. We build three deterministic simulators with exact gold state and seed-only randomness. **GraphNav** tests navigation through room-and-door graphs with keys, switches, and decoys. **ToolDAG** tests typed tool-call planning under a growing variable namespace. **StatefulPuzzle**, the selected confirmatory environment, asks the agent to move objects across rooms, containers, and slots through ordered subgoal chains. In StatefulPuzzle, SC counts the jointly maintained rooms, containers, and items, while DD counts the preconditions tying each subgoal to the current world state. The appendix gives formal definitions and reproducibility details.

Env	mini	haiku	match	min max $\Delta\hat{p}$
GraphNav	hump	u-shape	X	—
ToolDAG	flat	monotone	X	0 pp
StatefulPuzzle	mono	mono	✓	30 pp

Table 1: Pilot environment selection from one-dimensional DD sweeps ($n=10$ per level and model). The confirmatory grid is run on StatefulPuzzle because it is the only simulator whose pilot curve is monotone for both model tiers, has matching cross-model shape, and clears the 20-point minimum-drop criterion.

Trigger-environment selection. The trigger environment is fixed by a pilot-guided rule locked before any agent evaluation: choose the environment whose one-dimensional DD sweep ($n=10$ per level, two model tiers) satisfies (a) per-model monotonicity, (b) cross-model shape agreement, and (c) $\min(\max \Delta\hat{p}) \geq 20$ pp. The rule is deliberately closer to benchmark construction than to model selection: it chooses the simulator whose stress axis produces a readable, monotone diagnostic surface, a practice common in controlled planning benchmarks and long-horizon agent rollout suites (Valmeekam et al., 2022; Fang et al., 2026; Zhao et al., 2026). GraphNav fails (b); ToolDAG fails (c). StatefulPuzzle satisfies all three and is selected (Table 1).

Stress grid. The confirmatory grid covers $SC \in \{5, 10, 20, 40\}$ and $DD \in \{1, 2, 4, 6\}$ with secondary axes fixed at Regime III (horizon $T=40$, branching factor 4, clean observation, static mutation). Each of 16 cells contains $n=100$ episodes drawn from 10 archetypes $\times$ 10 instance variants with sha256-derived seeds (1,600 unique seeds, zero collisions).

Agent architecture. A three-call loop (Figure 2) runs identically at every step. The **Planner** proposes the next action and its expected state changes; the **Updater** rewrites the explicit working memory after the simulator step; and **Self-Diag** records a non-blocking validity judgment. The separation between acting, memory update, and self-checking is inspired by ReAct-style action loops, Reflexion-style self-diagnostics, and structured memory architectures (Yao et al., 2023b; Shinn et al., 2023; Chen et al., 2023; Packer et al., 2024). The confirmatory grid uses claude-haiku-4-5 at temperature 0 (Anthropic, 2025). Schema validation, retry policy, and budget triggers are reported in the supplementary material.

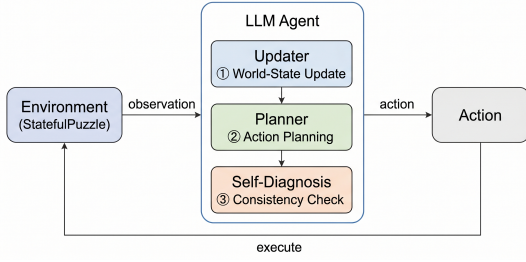


Figure 2: Three-call agent loop used in every episode. The Planner proposes an action from structured memory, the Updater rewrites the explicit world-state memory after the simulator step, and Self-Diag records a non-blocking validity judgment. Because the simulator exposes gold state after each action, the evaluator can separate representational failure from action failure while SC, DD, and the secondary axes are controlled.

4.1 Acceptance Criteria

The previous analysis asks two questions. First, does the grid contain an adjacent-pair cliff large enough to rule out a smooth difficulty curve? Second, when collapse occurs, do the world-state and action-validity traces reveal a stable temporal relationship? The cliff criterion passes on the confirmatory grid. The original symmetric synchrony criterion is too strict: the two failures are not simultaneous. The per-step trace instead gives the more informative mechanism, world-state collapse first and plan collapse later. The exact test definitions and the locked analysis list are reported in the appendix.

5 Results

5.1 Abrupt Transition on the Confirmatory Grid

The confirmatory grid in Figure 3 is the central empirical object of the paper: a phase diagram of world collapse rather than a smooth difficulty curve.

Structure. At low state cardinality, the agent solves the task across the dependency range. At high state cardinality, it collapses even when dependencies are sparse. The middle row contains the transition: increasing dependency density moves the agent through the critical band. This pattern is hard to reconcile with a one-dimensional difficulty curve. The secondary axis matters near the critical region, but once the world model has collapsed, adding dependencies no longer changes the

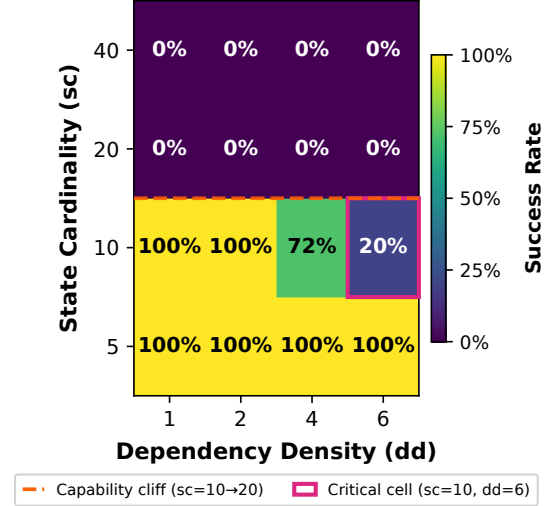


Figure 3: Confirmatory StatefulPuzzle success-rate heatmap for claude-haiku-4-5. The grid separates into a solved plateau, a narrow transition band, and a collapse floor. The sharpest boundary is along SC; within the transition band, larger DD moves the operating point toward collapse.

Table 2: Full confirmatory grid behind Figure 3. Each cell reports the success rate $\hat{p}$ over $n=100$ episodes. Bold cells form the solved plateau, italic cells are the transition band, and plain cells are the collapse floor. The table makes the core geometry explicit: the boundary is not a slow diagonal decline, but a plateau that falls sharply once state load crosses capacity.

SC	dependency density (DD)			
	dd=1	dd=2	dd=4	dd=6
5	1.00	1.00	1.00	1.00
10	1.00	1.00	<i>0.72</i>	<i>0.20</i>
20	0.00	0.00	0.00	0.00
40	0.00	0.00	0.00	0.00

outcome. The previous cliff criterion passes; the numerical tests are reported in the appendix.

The table and cross-sections are important because they show two views of the same object. Figure 3 shows the phase diagram globally, Table 2 shows that the plateau and floor are stable at the cell level, and Figure 4 shows that the decisive drop is localized along SC. Dependency density does not by itself create a long smooth decay; it changes where the agent sits relative to the state-capacity boundary.

World-state collapse comes first. The per-step trace explains why the heatmap has this shape. The world-state estimate loses fidelity before the planner’s actions become invalid. The original syn-

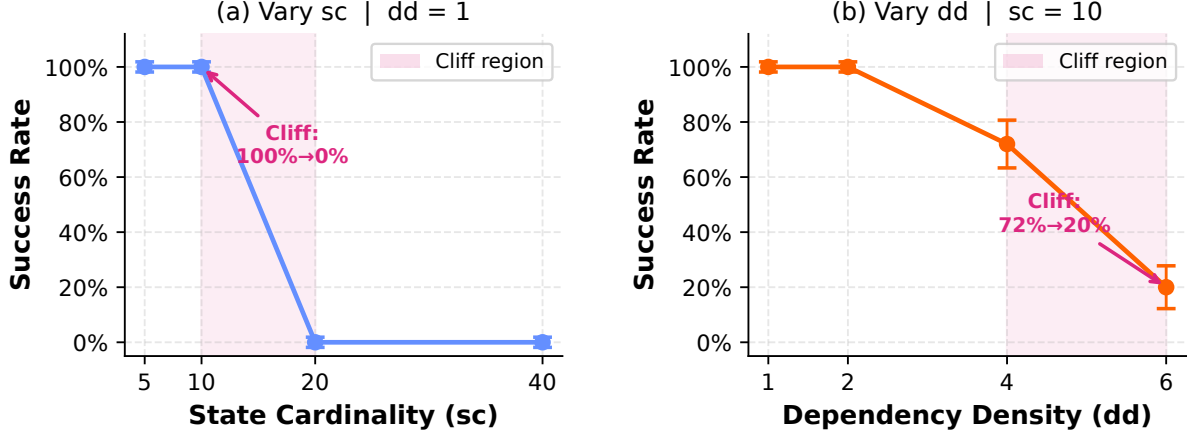


Figure 4: One-dimensional cross-sections of the confirmatory grid. The SC cross-sections expose the adjacent $SC=10 \rightarrow 20$ cliff at fixed DD. The DD cross-section at $SC=10$ shows that dependency density matters mainly near the boundary, where it moves the operating point through the transition band.

chrony criterion looked for simultaneous collapse and therefore fails; the data instead show an asymmetric sequence. This is the crucial mechanism: the plan fails because it is conditioned on a world representation that has already crossed its critical boundary. The appendix gives the locked onset analysis.

Temporal mechanism details. The paired collapsed episodes give a sharper picture than the original synchrony test. If collapse were a single undifferentiated event, world-state failure and invalid action would appear at the same step. Instead, the lag distribution is concentrated just before action failure: most paired episodes show the world-state estimate failing first, the median lead is two steps, and no paired episode shows a strict plan-first collapse. The symmetric synchrony rule captures less than half of the paired collapses because the modal event is just outside the simultaneous window. Thus the negative result on symmetric synchrony is not a weakness of the mechanism; it corrects the mechanism. The agent does not lose state and plan at the same instant. It first loses the state, then acts from the wrong state.

This temporal ordering also explains why final success alone is an insufficient diagnostic. A trajectory may look coherent until the first invalid action, but the representational error has already occurred. Per-step state instrumentation is therefore not a convenience feature of the benchmark; it is what makes the causal ordering visible.

Contrast with gradual drift. The canonical drift baseline (Lee, 2026) predicts a smooth profile as

stress increases. Here the doubled grid puts solved and failed regimes directly adjacent, while the fine scan in Figure 7 resolves a narrow transition rather than a long tail. The collapse floor is also insensitive to DD: once state load exceeds capacity, denser dependencies cannot make the already-failed world much worse. This combination—sharp cliff, narrow transition, and flat floor—is the empirical signature of world collapse. It extends the phase-transition view of static LLM outputs (Nakaishi et al., 2024) to sequential world-state dynamics.

5.2 Cross-Model Robustness

We next ask whether the boundary is an idiosyncrasy of one checkpoint or a capability-dependent property. The gpt-4o-mini grid in Figure 5 and Table 3 uses the same environment and agent harness as the confirmatory run. GPT-4o and Llama-3 70B provide additional cross-platform probes, summarized in Table 8. The probes cite the corresponding model releases or system cards: claude-haiku-4-5 (Anthropic, 2025), gpt-4o-mini (OpenAI, 2024a), GPT-4o (OpenAI, 2024b), and Llama-3 70B (Grattafiori et al., 2024).

The important result is not the cell-level numbers, but the preservation of shape. The easy cells remain easy, the high-SC cells still expose a floor, and the intermediate cells form a shifted transition band. In other words, changing the model translates the phase boundary rather than turning world collapse into smooth drift. Table 3 is useful precisely because it shows where that translation happens: the $SC=10$ row is no longer a near-binary cliff, while the higher-SC cells still reveal the same

Cross-Model Boundary Translation

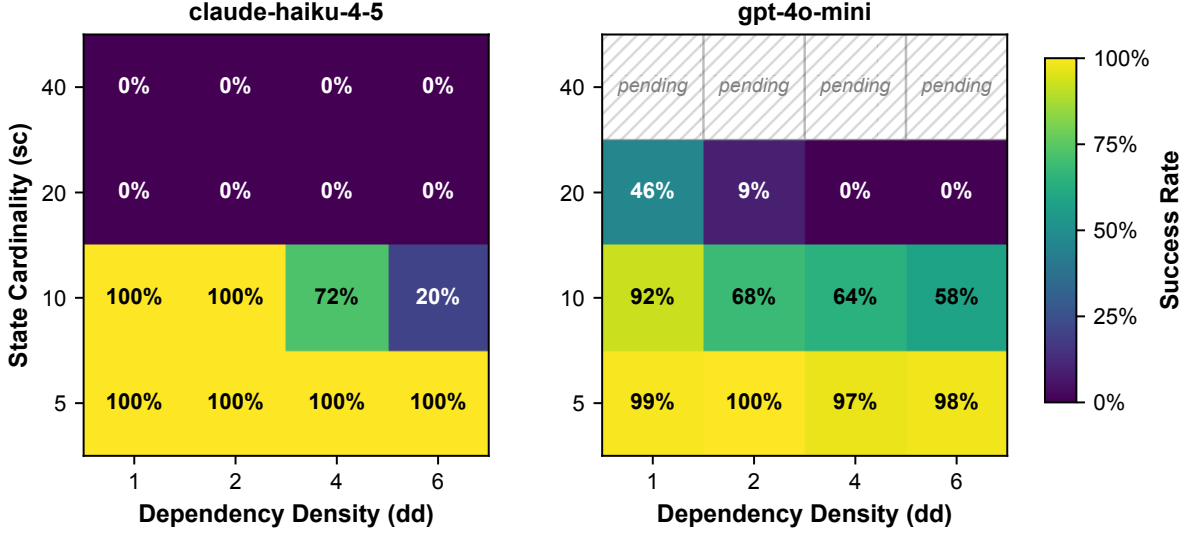


Figure 5: Cross-model comparison on the StatefulPuzzle grid. *Left:* claude-haiku-4-5 confirmatory grid. *Right:* gpt-4o-mini grid on the same stress axes. The qualitative phase structure is preserved—plateau, transition band, and floor—but the boundary moves. GPT-4o and Llama-3 70B probes are summarized in Table 8.

Table 3: Completed gpt-4o-mini cells used in Figure 5. The easy SC=5 row remains on the plateau, the SC=10 row becomes the transition band, and the SC=20 row approaches the floor as DD increases. This is the same phase-diagram shape as claude-haiku-4-5, shifted to a different boundary location.

SC	DD	n	$\hat{p}$
5	1	100	0.990
5	2	100	1.000
5	4	100	0.970
5	6	100	0.980
10	1	100	0.920
10	2	100	0.680
10	4	100	0.640
10	6	100	0.580
20	1	125	0.464
20	2	146	0.089
20	4	114	0.000
20	6	100	0.000

capacity limit.

GPT-4o corner cells. GPT-4o gives the same qualitative message. It lifts difficult interior cells, but still encounters a cliff as state cardinality increases. Capability moves the operating point and delays collapse; it does not remove the need to measure the boundary.

The corner probes sharpen this point. GPT-4o solves the easy corner and lifts the hardest SC=10 corner, but it does not make the SC=20 row uni-

formly reliable. Llama-3 70B, by contrast, sits below the observed boundary in the probed cells, with the endpoint-determinism caveat noted in the limitations. The comparison is therefore best read as a translation of the operating boundary under model and harness, not as a universal capability leaderboard.

The cross-model evidence is a capability gap acting on the *same* phase structure: every qualitative feature (monotone descent, plateau and floor, sharp intermediate-SC drop) carries over to mini and GPT-4o. Llama-3 70B is treated separately as a lower-bound probe with an endpoint-determinism caveat. The robust finding is boundary translation, not disappearance of the boundary.

5.3 Cross-Environment Considerations

The trigger-selection rule in Table 1 selected StatefulPuzzle because it was the only pilot environment whose curves were monotone and comparable across model tiers. GraphNav and ToolDAG are useful negative pilots: they show that not every controllable environment automatically produces a clean phase diagram. We therefore make the strong claim within StatefulPuzzle and treat cross-environment replication as a separate question requiring environment-specific instrumentation.

Table 4: Full secondary-axis ablations around the transition-zone backdrop: SC=10, DD=6, baseline horizon and branching, clean observations, and static mutation. The table supports the role assignment in Figure 6: horizon is enabling, branching is an intended null, observation is a visibility gate, and mutation remains descriptive.

Axis	Level	$\hat{p}$
T	$T=10$	0.000
	$T=20$	0.000
	$T=40$	0.120
	$T=80$	0.320
Branching	=2	0.160
	=4	0.200
	=8	0.160
	=16	0.200
Obs. noise	clean	0.400
	conflict	0.160
	distractor	0.120
	partial	0.000
Mut. rate	static	0.080
	low	0.320
	medium	0.240
	high	0.080

5.4 Single-Axis Ablations

The phase diagram should not be a disguised proxy for a single secondary axis. We therefore vary horizon, branching, observation mode, and mutation rate one at a time around a transition-zone backdrop. Figure 6 summarizes the response curves, and Table 4 reports the corresponding cells. The backdrop matters: if the ablations were run deep inside the plateau, every axis would look harmless; if they were run deep inside the floor, every axis would look irrelevant. We place them near the boundary so that a genuine alternative driver has room to express itself.

T (enabling). Horizon determines whether the agent has enough interaction budget to finish, but it does not itself define the world-model boundary. This generalizes context-budget accounts (Liu et al., 2024; Chung et al., 2025; Zhou et al., 2025; Luo et al., 2025): sufficient T is necessary but not sufficient.

Branching (intended null). Branching is nearly flat in this backdrop. The main cliff therefore cannot be explained as a hidden search-tree blow-up.

Observation noise (visibility gate). The observation modes separate by whether the true state remains visible. Partial observation hides state slots from the Updater; distractor and conflict modes

Table 5: Cliff’s δ for each ablation level against its backdrop. The negligible branching effects support the search-tree null; the observation effects are larger because partial observability removes state slots from the updater; mutation effects remain small to medium and are treated as descriptive. Codes denote negligible (n), small (S), and medium (M) effects.

axis	levels vs. base	δ
T	10/20/80 vs. 40	-0.12, -0.12, +0.20 n, n, S
Branching	2/8/16 vs. 4	-0.04, -0.04, +0.00 n, n, n
Obs. noise	partial/distractor/conflict vs. clean	-0.40, -0.28, -0.24 M, S, S
Mut. rate	low/medium/high vs. static	+0.24, +0.16, +0.00 S, S, n

leave the relevant state available amid noise. *Observability*, not nominal noise, couples this axis to the transition (Liu et al., 2024; Ponnusamy et al., 2025).

Mutation rate (descriptive). Mutation rate shows a suggestive low-rate hump, which we retain only as a candidate informativeness modulator.

Synthesis. The four axes play different roles—enabler, null, visibility gate, and descriptive modulator. That diversity is exactly what a single scalar “effective difficulty” account would struggle to explain. The dominant structure remains the two-dimensional (SC, DD) boundary. Table 5 provides a compact effect-size view of the same conclusion: branching stays negligible, while observation changes matter when they remove the state needed by the Updater.

5.5 Critical-Point Localisation

The doubled grid shows that the main cliff lies between SC=10 and SC=20. We refine that picture with two scans in Figure 7: one along SC, where the boundary should live, and one along horizon, where a context-budget account would expect a similar threshold. This paired design is a negative control as much as a localization tool. A phase-transition story predicts a narrow crossing along the structural state axis. A horizon-budget story predicts that the same kind of crossing should appear when the interaction length is swept at fixed structure. Only the first prediction is borne out.

SC-Fine. The SC scan turns the coarse cliff into a localized critical region. Values just below the boundary remain on the plateau, values just above it fall rapidly to the floor, and the crossover lies near $SC^* \approx 13.5$. Thus the doubled-grid cliff is not an artifact of sparse sampling; it is a narrow boundary visible at unit-integer resolution (Table 6).

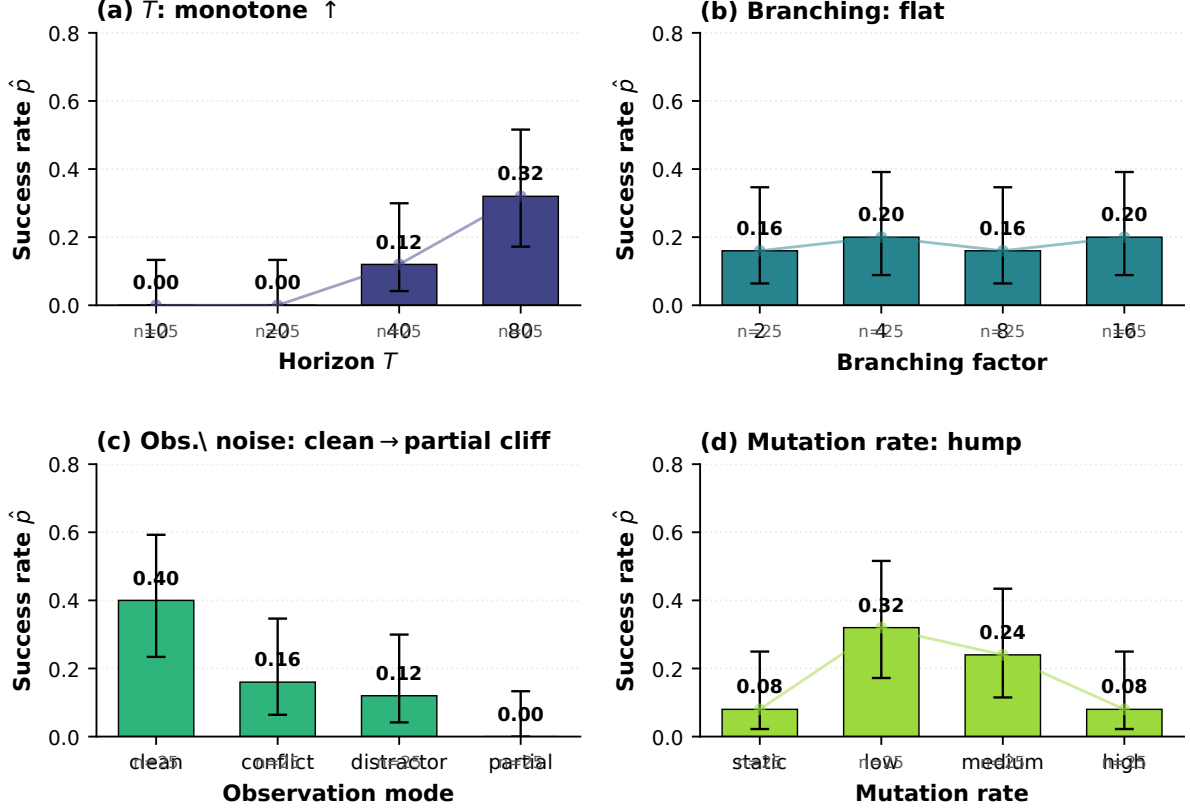


Figure 6: Single-axis ablations around a transition-zone backdrop. (a) Horizon acts as an enabler: too few steps prevent success, but more steps do not define the phase boundary. (b) Branching is the intended null, showing that the main cliff is not a search-tree artifact. (c) Observation mode acts as a visibility gate: hiding state slots is more damaging than adding visible distractors. (d) Mutation rate is descriptive rather than a confirmed driver.

Table 6: SC-Fine scan at fixed DD=1 for claude-haiku-4-5, with confirmatory-grid endpoints included as references. The crossover is bracketed by SC=13 and SC=14; by SC=15, the system is already on the collapse side of the boundary.

SC	succ	n	$\hat{p}$
10*	100	100	1.000
11	50	50	1.000
12	50	50	1.000
13	39	50	0.780
14	26	50	0.520
15	1	78	0.013
16	1	50	0.020
17	3	50	0.060
18	0	50	0.000
19	0	50	0.000
20*	0	100	0.000

* Confirmatory-grid endpoint.

T-Fine. The T scan behaves differently. Inside the transition-zone backdrop, additional horizon does not reveal a comparable critical point. Horizon remains an enabling resource, but the phase boundary is governed by the structural axes. We

therefore retract the earlier expectation of a resolved T^* and treat T as a modulator rather than a driver (Table 7).

5.6 Boundary Translation Across Models

Combining the fine scan, the gpt-4o-mini grid, and the cross-platform probes, Table 8 reads the boundary as a model-dependent capacity marker. The table is qualitative by design: the claim is boundary translation, not a definitive ranking among all model families. This distinction keeps the interpretation conservative. The exact crossing depends on both the model and the harness, but the relevant invariant is the shape of the surface: models move from one operating point to another on the same structural diagram.

Reading the table. Llama-3 70B sits below the boundary in the tested cells, with an endpoint-determinism caveat discussed in the Limitations. Claude-haiku-4-5 gives the cleanest localized boundary. The OpenAI checkpoints shift several difficult cells upward, but still encounter a state-

Critical Point Localisation

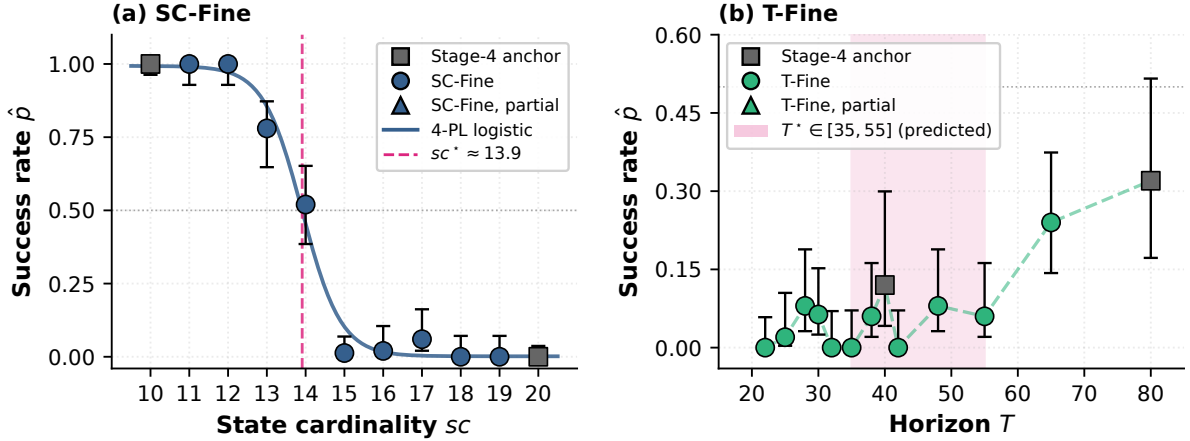


Figure 7: Critical-point scans for claude-haiku-4-5 on StatefulPuzzle. (a) SC-Fine resolves the doubled-grid cliff by sweeping integer sc values between the plateau and the floor; the crossover lies near $sc^* \approx 13.5$. (b) T-Fine remains flat inside the same operating region, so no analogous T^* is resolved.

Table 7: T-Fine at fixed $(sc, DD) = (10, 6)$ for claude-haiku-4-5, with the coarse ablation reference cells marked by *. Unlike SC-Fine, this interior horizon scan does not reveal a monotone critical boundary; it supports the interpretation of T as an enabling modulator.

T	succ	n	$\hat{p}$
22	0	62	0.000
25	1	50	0.020
28	4	50	0.080
30	4	63	0.063
32	0	51	0.000
35	0	50	0.000
38	3	50	0.060
40*	3	25	0.120
42	0	50	0.000
48	4	50	0.080
55	3	47	0.064
80*	8	25	0.320

cardinality cliff. The comparison therefore supports the same conclusion as Figure 5: capability translates the boundary.

Interpretation. We do not interpret the table as a universal capability leaderboard. Architecture, endpoint behavior, and harness compatibility can all move the observed boundary. What survives those caveats is the phase-transition form itself. Table 9 gives the crossing-location readout where the crossing is bracketed, and Table 10 shows the complementary cellwise effects. Together they separate two claims: models can move specific operating points substantially, but degenerate floor cells leave little room for capability to appear.

Table 8: Cross-model boundary readout on the $DD=1$ row. Stronger or better-supported models move the collapse boundary outward, but the plateau–transition–floor shape remains. Llama-3 70B is included as a cross-platform lower-bound probe.

model	boundary readout	interpretation
Llama-3 70B	below tested low- sc boundary	lower-bound probe
claude-haiku-4-5	$sc^* \approx 14$	localized cliff
gpt-4o-mini	high-teens boundary	boundary shifted outward
GPT-4o	between low and mid sc	strong interior lift, cliff remains

Table 9: Cross-model sc^* readout on the $DD=1$ row. The table gives a compact quantitative summary of the qualitative boundary translation in Table 8. Llama-3 70B does not bracket the 50% crossing in the populated cells and is therefore reported as a lower-tier probe.

model	sc^*	readout
claude-haiku-4-5	14.04	localized boundary
gpt-4o-mini	18.94	shifted outward
GPT-4o	15.56	strong interior lift
Llama-3 70B	— (< 10)	below tested low- sc boundary

5.7 T as Modulator, Not Driver

The T-Fine scan strengthens the sc -axis interpretation. If the cliff were merely a reparameterized horizon effect, an interior horizon scan should reveal a comparable boundary. Instead, the scan remains flat inside the collapse band. Horizon still matters as an enabling resource—too little time can prevent success—but it does not set the phase boundary. The boundary is structural: (sc, DD) determine where the world model fails, while T determines whether the agent has enough steps to exploit the state it

Table 10: Cross-model Cohen’s h across shared cells. The large effects show that model choice can substantially move particular operating cells, while the degenerate floor cells show that once collapse has occurred, there is little remaining room for capability to express itself. Codes denote negligible (n), small (S), medium (M), and large (L) effects.

pair	(10, 1)	(10, 6)	(20, 1)
haiku–mini	+0.57 M	−0.80 L	−1.47 L
haiku–GPT-4o	+0.00 n	−1.29 L	−0.64 M
haiku–Llama-3	+2.21 L	+0.93 L	+0.00 n
mini–GPT-4o	−0.57 M	−0.48 S	+0.83 L
mini–Llama-3	+1.64 L	+1.73 L	+1.47 L
GPT-4o–Llama-3	+2.21 L	+2.21 L	+0.64 M

can still maintain.

6 Discussion

The results point to a simple failure model. A long-horizon agent can plan only while its working world remains coherent. When state cardinality and dependency density cross the critical region, the planner does not gradually become less clever; it acts on the wrong world. This reframes what the heatmap is measuring. The axes are not surface difficulty knobs in the ordinary sense. They control how much of the world must be kept addressable and how often future decisions must bind back to that representation. The collapse is therefore a failure of maintained structure, not just a failure of final-step reasoning.

Cognitive precedence: world state before plan.

The strongest mechanistic evidence is temporal. World-state collapse appears before plan collapse. This ordering is not a minor diagnostic detail: it says which component fails first. Once the Updater loses the relevant state, the Planner receives a stale or inconsistent view and action validity follows it down. This is why the original symmetric synchrony test was the wrong story; the failures are coupled, but not instantaneous.

Phase-transition geometry. Figure 1 summarizes the geometry. The (SC, DD) plane has a solved regime, a transition zone, and a collapse floor. State cardinality sets the compositional load; dependency density controls how tightly subgoals are coupled to that state. The boundary moves leftward as dependencies become denser, but the floor remains flat once state load is beyond capacity. This is the feature that distinguishes the result from smooth drift (Lee, 2026) and from metric-only

accounts of apparent emergence (Schaeffer et al., 2023).

Capability and scaffolding. The cross-model probes suggest that capability translates the boundary rather than changing the shape of the diagram. This matters for agent design. If the first failing component is the world model, then simply asking the same planner to reason harder is unlikely to solve the problem. More promising interventions externalize or decompose state: structured memories (Packer et al., 2024; Park et al., 2023; Sumers et al., 2024; Arslan, 2026; Shi et al., 2025), planner–solver hybrids (Kambhampati et al., 2024; Liu et al., 2023a; Silver et al., 2024), or explicit world-state representations (Zhu et al., 2026). The aim is to push SC^* outward by supporting the representation that fails first. This also changes how scaffolds should be evaluated. A memory module should not only improve final success; it should delay τ_W . A planner should not only recover after invalid actions; it should avoid conditioning on stale state. A benchmark should therefore report where the boundary moves, not only whether the mean score improves.

Cross-disciplinary link and implications. The evidence resembles physical phase-transition phenomenology: a sharp control-axis boundary, flat plateau and floor, a model-specific critical location SC^* , a narrow transition band, and a precursor signal. The same vocabulary has been productive across deep learning (Wei et al., 2022; Schaeffer et al., 2023; Nakaishi et al., 2024; Power et al., 2022; Olsson et al., 2022; Belkin et al., 2019; Bahri et al., 2020; Roberts et al., 2022). We do not claim a thermodynamic-limit phase transition, but the analogy is useful: it tells us to look for boundaries, precursors, and model-specific critical locations. For evaluation, this means averaged benchmark scores are not enough. A benchmark that does not vary state load and dependency load independently can average over the cliff and make a threshold look like ordinary drift. The practical implication is to complement realistic agent benchmarks with small controlled stress grids. Realistic tasks tell us whether an agent is useful; controlled grids tell us why a useful-looking agent fails. The two are not substitutes. The phase diagram gives a compact diagnostic for one failure mode that broad benchmarks can otherwise hide.

7 Conclusion

Long-horizon LLM agent collapse is better described as a world-model phase transition than as ordinary gradual drift. When state cardinality and dependency density cross a critical region, the agent first loses the represented world and only then loses valid action. Fine scans localize this boundary, cross-model probes show that stronger models shift it rather than erase it, and secondary-axis ablations show that horizon, branching, observation, and mutation play distinct supporting roles. The practical lesson is direct: world-model capacity is a measurable, model-specific bottleneck. Evaluations that only average final success over naturalistic tasks can hide this boundary; reliable long-horizon agents require stress grids, per-step state instrumentation, and scaffolds that support the world representation before the planner fails. The broader message is that agent evaluation should measure the state the agent thinks it is acting in, not only the action it finally takes.

Limitations

Controlled scope. The confirmatory claim is made inside StatefulPuzzle, the environment selected by the locked pilot rule. GraphNav and ToolDAG are useful negative pilots, but they are not confirmatory replications. The paper therefore establishes a clean phase-transition testbed rather than a universal claim about every agent environment.

Synchrony was the wrong mechanistic form. The locked synchrony criterion asked for simultaneous world-state and plan collapse. The data instead support an asymmetric precedence story: world state fails first. We make that correction explicit in the main text. A separate calibration of the serial Planner/Updater/Self-Diag pipeline would further separate architectural lag from stress-dependent collapse lag.

Secondary axes are not equally established. Horizon, branching, observation mode, and mutation rate are included to rule out simple alternatives, not to exhaustively model every stressor. The mutation-rate pattern is treated as descriptive, and the branching null only rules out large search-tree explanations under this backdrop.

Finite-grid phase transition. We do not claim a thermodynamic-limit phase transition or estimate

critical exponents. The claim is operational: under the tested grid, the success surface has a plateau, a narrow transition, and a collapse floor. Finer and larger grids would be needed to study scaling laws.

Platform heterogeneity. The model probes span different providers and serving interfaces. Endpoint determinism, seed control, and chat formatting are not identical across platforms, so the cross-model table should be read as boundary translation evidence rather than as a clean capability leaderboard.

Architecture and harness narrowness. The model set spans several serving stacks, but the agent architecture is fixed to a three-call Planner/Updater/Self-Diag loop with structured working memory. This makes the mechanism observable, but it also means the observed cliff may be a property of the model–harness pair. Running the same grid with raw memory, summarized memory, or external symbolic state would test whether the boundary moves or changes shape.

Naturalistic-benchmark sanity check. We do not test on WebArena (Zhou et al., 2024), SWE-Bench (Jimenez et al., 2023), or TravelPlanner (Xie et al., 2024a). Whether the (SC, DD) cliff lifts to naturalistic benchmarks where these axes are confounded with horizon, branching, and environment heterogeneity is left to future work; the present paper trades ecological breadth for the per-step ground truth and orthogonal stress control needed to test the phase-transition claim mechanistically.

Discretized axes. The (SC, DD, T , branching) axes are sampled at engineering-tractable doubling levels with integer-resolution refinements inside the transition windows; we do not estimate critical exponents or scaling laws. Reported boundary locations are interpolated from the integer grid. Observation noise and mutation rate are categorical, so those findings should be read as ordered category-level effects rather than continuous thresholds.

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A Proof of the Grid-Bracketing Criterion

Proposition 1 (Grid bracketing under monotonicity). *Fix dependency density d and held-fixed factors z . Suppose $p_\theta(s, d; z)$ is non-increasing in s on an interval $[s_i, s_{i+1}]$ and that $p_\theta(s_i, d; z) > 1/2 > p_\theta(s_{i+1}, d; z)$. Then every right-continuous version of the critical location $s_\theta^*(d; z) = \inf\{s : p_\theta(s, d; z) \leq 1/2\}$ lies in $[s_i, s_{i+1}]$. If the observed grid has no cell with success estimate in $[\eta, 1 - \eta]$ for $0 < \eta < 1/2$, then the observed transition width at that grid resolution is zero under the point-estimate criterion.*

Proof. Because $p_\theta(s_i, d; z) > 1/2$, the set $\{s : p_\theta(s, d; z) \leq 1/2\}$ cannot begin before s_i under monotonicity. Because $p_\theta(s_{i+1}, d; z) < 1/2$, the same set is nonempty by s_{i+1} . Its infimum therefore lies in $[s_i, s_{i+1}]$. For the grid-width statement, the point-estimate transition width counts grid points whose estimated success lies inside the middle band $[\eta, 1 - \eta]$. If no grid point satisfies that predicate, the count is zero by definition. The proposition is used only to justify the operational bracketing of a finite-grid crossover; it does not assert a thermodynamic-limit phase transition. $\square$

B Environment Specifications

All three environments expose the same transition contract. An episode is initialized from a configuration and seed, advances one action at a time, and exposes an exact gold state together with oracle labels for action validity and error type. The gold state is a structured record of all world facts, and no environment-side randomness is introduced beyond the seed.

Graph Navigation (GraphNav). Agents navigate a room-and-door graph under key, switch, and decoy constraints. SC is the node count; DD is the number of preconditions to unlock each door. Error labels: nonexistent_edge, missing_key, stale_inventory.

Tool-DAG (ToolDAG). Agents execute a directed acyclic graph of typed tool calls, maintaining a variable namespace. SC is the number of active variables; DD is the number of typed input arguments per tool. Error labels: missing_argument, skipped_dependency, fabricated_result.

Stateful Puzzle (StatefulPuzzle). Agents manipulate objects across rooms, containers, and item slots through ordered subgoal chains. SC counts rooms+containers+items; DD is the number of preconditions per subgoal. Concretely: $sc=5 \rightarrow 2$ rooms, 1 container, 2 items, 2 subgoals; $sc=10 \rightarrow 3$ rooms, 3 containers, 4 items, 3 subgoals; $sc=20 \rightarrow 5$ rooms, 6 containers, 9 items, 4 subgoals; $sc=40 \rightarrow 8$ rooms, 12 containers, 20 items, 6 subgoals. Action space: go, take, put, open/close, use, combine, activate, examine, finish_subgoal, noop. Error labels: precondition_violation, object_location_error, stale_room_state.

C Stress-Regime Specification

The five *world regimes* used in our experimental design fix all stress axes that are not under direct manipulation in a given study. Each regime is a tuple over six axes: state cardinality (SC), dependency density (DD), horizon T , branching factor, observation noise mode, and mutation rate. Table 11 lists the fixed backdrops used across the confirmatory grid, ablations, and fine scans.

Table 11: Five world regimes used as fixed backdrops across the study. Regime III is the confirmatory-grid backdrop; the single-axis ablations probe one axis at a time from Regime III; SC-Fine and T-Fine probe along sc and T respectively from Regime III.

Regime	SC	DD	T	branch	obs. noise	mut.
I (Trivial)	5	1	40	4	clean	static
II (Moderate)	10	2	40	4	clean	static
III (Coupled)	—	—	40	4	clean	static
IV (Adversarial)	20	4	40	4	partial	low
V (Extreme)	40	6	80	8	conflict	medium

Regime III holds horizon, branching, observation noise and mutation rate at their baseline values

and sweeps (SC, DD) on the main 4×4 stress grid.

D Memory Representation

The three-call loop can be coupled to several memory representations: raw step transcripts, rolling natural-language summaries, or an explicit structured world state maintained by the Updater. All experiments reported in this paper use the structured-memory representation, because the purpose is to measure when an explicit world model remains stable under controlled stress. Transcript-only and summary-only variants are reserved for a separate memory-architecture study.

E Agent Loop Interface

The structured-memory agent communicates through three typed interfaces. The Planner proposes exactly one action, states the preconditions it believes support that action, predicts the action’s local effects, and reports a scalar self-rating. The simulator receives only the proposed action; the self-rating is recorded but never used to rescue or veto the choice.

The Updater is responsible for the explicit world state. Given the latest observation, it adds newly true facts, removes facts made stale by the transition, and emits the complete state that will condition the next Planner call. Self-Diag then provides an independent judgment of the proposed action: whether it appears valid, which preconditions appear missing, and whether replanning would be preferred. This diagnostic is observational rather than interventional, so the environment remains the sole arbiter of success or failure. Malformed interface outputs are repaired by a bounded retry rule and otherwise mapped to deterministic defaults, ensuring that parsing failures do not become an unmodeled source of stochasticity.

F Budget and Stopping Rules

The evaluation uses fixed resource ceilings to prevent pathological episodes from dominating the grid. Each episode has an output-token ceiling of 80k, a wall-time ceiling of 30 minutes, and a maximum of three repair attempts for malformed interface outputs. The fallback-rate trigger was fixed before the confirmatory run and used only as a safeguard against an invalid measurement regime. In practice, fallback was rare and did not determine any headline effect, so the ceilings should be read

as run-control safeguards rather than experimental variables.

G Evaluator and Metrics

The evaluator compares the agent state and action against the environment’s deterministic gold oracle after each transition. Five per-step quantities are recorded:

- world-state accuracy: $\text{Jaccard}(\hat{W}_t, W_t^*)$ between agent-maintained world state and gold;
- action validity: gold preconditions for the chosen action all hold in W_t^* ;
- world consistency: $\hat{W}_t$ satisfies the environment’s invariant predicates;
- dependency correctness: required-precondition set declared by the Planner matches gold;
- self-check accuracy: agreement between Self-Diag verdict and gold action-validity verdict.

Episode-level metrics include final success (gold goal predicate satisfied at episode end), collapse onset τ_o (first step of a 3-of-next-5 bad window), and collapse type (*world-state*, *action-validity*, *self-check*, or *compound*).

H previous Decisions

The following decisions were fixed before confirmatory data collection. The lock separates the primary phase-transition claim from later scans that help interpret the geometry of the transition surface.

- Goal G1 acceptance criterion: G1a (cliff existence) AND G1b (multi-metric synchrony); both required for Goal G1.
- G1a primary statistic: Miettinen–Nurminen score test for $H_0: \Delta p \leq 0.30$ vs. $H_{\text{alt}}: \Delta p > 0.30$, one-sided, $\alpha = 0.01$.
- G1b synchrony statistic: per-metric one-sided Mann–Whitney U test on world-state accuracy, action validity, and self-check accuracy at the locked trigger pair, Holm-corrected at $\alpha = 0.01$, with Hodges–Lehmann $\hat{\Delta}_k \geq 0.20$ for each metric. *G1b did not pass; the operational mechanism claim therefore rests on the precedence analysis reported in the main text.*
- Stress grid: $\text{SC} \in \{5, 10, 20, 40\}$, $\text{DD} \in \{1, 2, 4, 6\}$; $n = 100$ episodes per cell.
- Backdrop axes: Regime III (Coupled).
- Collapse onset definition: 3-of-next-5 bad steps, where $\text{bad}(t) :=$

$\neg \text{action_valid} \vee \neg \text{world_consistent} \vee \neg \text{dependency_correct}$.

- Trigger-environment selection rule: the selection rule was fixed before the pilot sweep and then applied to choose the confirmatory environment.
- N-scaling diagnostic: $W_{\text{trans}}(d) := |\{N: 0.30 \leq \hat{p}(N, d) \leq 0.70\}|$ via point estimate.

I Statistical Analysis

The previous primary analysis tests the existence of the main cliff. Secondary axis tests ask which ingredients open or close the brittle regime: horizon, branching, observation mode, and mutation rate. Under both Bonferroni control at $\alpha = 0.01$ and Benjamini–Hochberg control at $q = 0.05$, the same substantive effects remain: the main cliff, the horizon enabling effect, and the observation-visibility effect. Branching is the intended null, and mutation is treated only as a descriptive modulation.

The later scans are not additional confirmatory hypotheses. SC-Fine localizes the state-cardinality boundary, T-Fine asks whether a comparable horizon boundary appears, and the cross-model probes read out how the same boundary geometry translates under different model–harness pairs. The abstract, introduction, and conclusion therefore restrict their claims to the primary family and to the qualitative replication of the phase-diagram geometry.

For the primary cliff test we use the Miettinen–Nurminen score statistic

$$T = \frac{(\hat{p}_1 - \hat{p}_2) - \delta_0}{\sqrt{\tilde{p}_1(1 - \tilde{p}_1)/n_1 + \tilde{p}_2(1 - \tilde{p}_2)/n_2}},$$

where $\delta_0 = 0.30$ and $(\tilde{p}_1, \tilde{p}_2)$ are the constrained-MLE proportions under $H_0: p_1 - p_2 = \delta_0$ (Miettinen and Nurminen, 1985). This one-sided test formalizes the claim that the low-stress and high-stress cells are separated by a practically large drop, rather than by a small monotone drift.

J N-Scaling Diagnostic

For each dependency-density column d , define

$$W_{\text{trans}}(d) = \sum_{N \in \{5, 10, 20, 40\}} I_N(d),$$

$$I_N(d) = \mathbf{1}\{0.30 \leq \hat{p}(N, d) \leq 0.70\}.$$

The previous point-estimate variant uses $\hat{p}(N, d)$ directly; cells satisfying the band condition count

toward W_{trans} . The diagnostic is not an additional acceptance test. Its role is descriptive: it compresses the finite grid into a statement about whether the transition region broadens or sharpens across dependency-density columns.

Three patterns

- Pattern A (Sharpening): $W_{\text{trans}}(d)$ is non-increasing in d and asymptotes to a constant width ≤ 1 .
- Pattern B (Constant): $W_{\text{trans}}(d)$ is constant in d .
- Pattern C (Broadening): $W_{\text{trans}}(d)$ is non-decreasing in d , consistent with a finite-size crossover.

The doubled confirmatory grid assigns no cell to the middle band by point estimate. SC-Fine then refines the $d=1$ column by resolving the middle band into the two integer locations $SC=13$ and $SC=14$. The refinement narrows the boundary at unit resolution without changing the previous verdict.

K Reproducibility

The simulator side is fully deterministic. Task seeds are derived from the tuple consisting of the grid cell, archetype, and instance identifier, yielding unique seeds across the confirmatory grid. Model calls use temperature zero, a fixed maximum response length, and the retry policy described above. The released materials include the deterministic environment oracles, prompts, per-cell traces, and the pre-registration record. The analysis code recomputes the grid summaries and the Miettinen–Nurminen test used for the primary cliff criterion.

L Cross-Platform Model Probes

The cross-model probes keep the structured-memory harness fixed and vary only the served model and provider interface. The primary grid uses claude-haiku-4-5 through the Anthropic Messages interface (Anthropic, 2025). The OpenAI probe uses gpt-4o-mini through Chat Completions (OpenAI, 2024a); the stronger OpenAI probe uses GPT-4o through Azure OpenAI (OpenAI, 2024b); and the open-weight probe uses Llama-3 70B Instruct through AWS Bedrock (Grattafiori et al., 2024). All probes use the same Planner, Updater, and Self-Diag prompts. Providers differ in whether they expose a deterministic seed parameter, so the cross-model comparison is interpreted

as a boundary-translation probe rather than a bit-identical replication.